

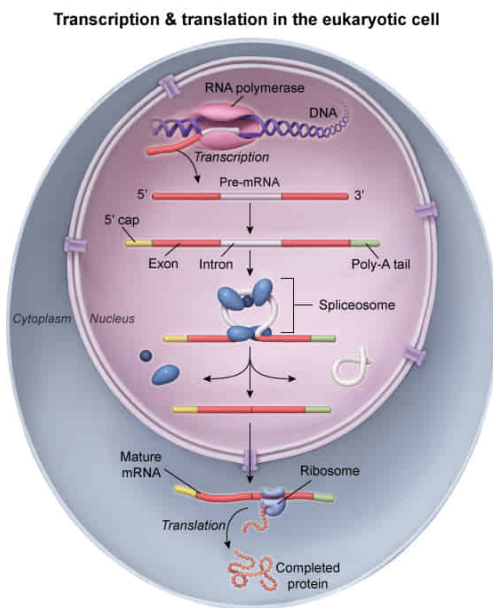
Plasmodium falciparum is a protozoan parasite known to cause malaria, an illness characterized by fever, chills, and flu-like symptoms. Which of the following is the most likely reason why *P. falciparum* would be classified as a eukaryotic organism?

- ☐ A. It is able to inherit genetic information from another individual. [9%]
- ✓ ☒ B. It has a spliceosome-mediated mechanism for intron splicing. [70%]
- ☐ C. It has a cell wall. [3%]
- ☐ D. It has a cell membrane. [15%]

Correct

69% answered correctly

Explanation:



Eukaryotic cells have distinct characteristics that distinguish them from prokaryotic cells. For example, membrane-bound organelles are unique to eukaryotic cells and allow for the compartmentalization of specific cellular processes (eg, transcription in the nucleus, aerobic respiration in the mitochondria).

In addition, eukaryotic cells have a unique mechanism for processing messenger RNA (mRNA). Eukaryotic mRNA is initially transcribed as pre-mRNA, containing both introns (noncoding sequences) and exons (coding sequences). Following transcription, the **spliceosome** (an RNA and protein complex) **removes introns** from the pre-mRNA molecule to form mature mRNA in a process known as **RNA splicing**. This mature mRNA is transported out of the nucleus and into the cytoplasm, where it is translated into protein. Although introns have been identified in a few prokaryotic organisms (eg, some archaea), spliceosome-mediated intron splicing is generally a process that is unique to eukaryotes.

Therefore, because **protozoa** (eg, *Plasmodium falciparum*) are **single-celled eukaryotic organisms**, they would possess all the *unique* characteristics of

eukaryotic cells, including a spliceosome-mediated mechanism for splicing introns from mRNA in the nucleus.

(Choice A) Both eukaryotic and prokaryotic organisms can inherit genetic information from other individuals. For example, in eukaryotic organisms, offspring produced sexually (ie, via **gamete fusion**) inherit genetic information (ie, alleles) from parental organisms. In prokaryotes, **conjugation** allows the transfer of genetic material from one cell to another. Therefore, the ability to inherit genetic information from other individuals is *not* unique to eukaryotes.

(Choice C) The cell wall is a rigid polysaccharide layer surrounding the cell membrane that provides protection and structural support in certain cell types. Most prokaryotic cells have cell walls, but only some eukaryotic cells (ie, plants, fungi) have them. Because a cell wall can be found in both prokaryotes and eukaryotes, it would not be a distinguishing characteristic for classifying *P. falciparum* as eukaryotic. In addition, protozoa lack cell walls.

(Choice D) Because both eukaryotic and prokaryotic cells are surrounded by a cell membrane, this cannot be used to classify *P. falciparum* as eukaryotic.

Educational objective:

Eukaryotic and prokaryotic cells are similar in that both can pass genetic information to other individuals (eg, via sexual reproduction in eukaryotes and conjugation in prokaryotes). In addition, both organism types are enclosed by a plasma membrane. All prokaryotes and some eukaryotes have cell walls; however, eukaryotic cells are unique in that they have membrane-bound organelles as well as a mechanism for intron splicing by the spliceosome.

Time spent:0 sec

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